Specification Document

# Securing XML Web Services: Credit Card Validation Service

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## 1. Summary of Requirements & Objectives

This specification outlines the development requirements for an interactive Credit Card Validation Service. The solution leverages Microsoft Visual Studio.NET, XML Web Services, and ASP.NET to enable secure, real-time credit card authentication and validation.

The primary objectives of this project focus on driving progress, robust security architectures, and system improvements across several key areas:

- Utilize Internet Information Services (IIS) authentication to manage access control to the XML web services.
- Develop client applications capable of passing credentials securely to authenticate with protected XML Web services.
- Implement ASP.NET authorization controls to rigorously limit access to underlying XML Web service resources.

## 2. XML Web Service Core Features & Methods

The core business logic of the service consists of an automated Credit Card Validation Process. A fully compliant solution must incorporate the following specific validation features:

- Provider Verification: Ensure that the credit card number has been issued by a supported, valid provider.
- Expiration Check: Validate that the card has not expired.
- Status Check: Ensure that the credit card has not been canceled by the appropriate card issuer.
- Credit Authorization: Verify that the cardholder has a sufficient credit limit to execute a transaction.

### Validation Criteria & Methods

Developers must implement specific validation methods tailored to the following primary credit card types: American Express (AMEX), VISA, and MasterCard. Each method must verify the correct prefix, validate the total card length, and apply the LUHN formula (Mod 10 check) to prevent random number entries.

| Credit Card Issuer | Valid Prefixes | Valid Length | Target Method |
|---|---|---|---|
| VISA | 4 | 13, 16 | ValidateVisa |
| MasterCard | 51, 52, 53, 54, 55 | 16 | ValidateMasterCard |
| AMEX | 34, 37 | 15 | ValidateAMEX |

## 3. Client Application Requirements

A desktop or console client application must be developed in C# to invoke the credit card validation process. This client application will act as the consumer of the secure XML Web Services, handling the submission of payload data and rendering the interactive service validation results back to the end-user.

## 4. Security & Authentication Infrastructure

To safeguard the web services against unauthorized use, an explicit authentication mechanism is required:

- **Form-Based and Basic Authentication:** Build a robust, form-based authentication subsystem within ASP.NET. This should preferably utilize an optimized database lookup mechanism to match credentials against valid registered users.
- **Client Integration:** Implement secure client-side validation routing that leverages an ASP.NET form-based login interface (e.g., logon.asp) directly from the external C# application wrapper before granting service invocation rights.

## 5. Extended Custom Features (Creative Initiative)

Developers are encouraged to apply creative professional judgment to build out further business methods derived from the high-level requirements. Recommended expansions include standalone endpoints for programmatic expiration evaluation, automated provider lookup routers, and secure card status blacklisting modules.